

OptiCrop: Brainstorming & Idea Prioritization

Problem Space

Farmers often choose crops based on tradition or guesswork rather than actual soil and climate suitability, leading to lower yields, wasted inputs (fertilizer, water), and inconsistent income.

Candidate Ideas Considered

Idea	Priority	Reason
ML-based crop recommendation from soil/climate data	High	Directly solves the core problem; data (Crop_recomm
Soil-image-based deficiency detection (computer vision)	Medium	Valuable but needs image data collection; larger sco
Market-price-based crop advisory	Low	Depends on external live market data feeds; out of s
Chatbot-based farmer assistant	Low	Nice-to-have interface layer, not core to the predictio

Selected Idea

Build a supervised ML model that recommends a crop from 7 measurable soil/climate inputs (N, P, K, temperature, humidity, pH, rainfall), plus lightweight exploratory analysis (seasonal grouping, clustering) to make the recommendation more interpretable, and serve it through a simple web form.